

AVINASH KUMAR

Lead Software Engineer

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Summary

Lead Software Engineer with **6+ years of experience** designing and building scalable distributed systems, microservices architecture, and payment infrastructure. Architected backend platforms processing **\$50M+ in monthly payment volume**, with deep expertise in **Python, Django, Celery, Redis, and AWS**. Proven track record of leading engineering teams and owning systems end-to-end, from architecture to production.

Professional Experience

Practice by Numbers — US-based Dental Healthcare SaaS (3,000+ practices globally)

Gurugram

Lead Software Engineer

07/2026 – Present

Senior Software Engineer

07/2024 – 06/2026

Software Engineer

02/2023 – 06/2024

- **Led a team of 7 engineers** in designing and delivering a microservices-based payments platform supporting credit and debit card payments across POS and online channels, as well as Pay by Bank (ACH), processing **\$50M+ in monthly payment volume**.
- Grew payment revenue by redesigning checkout workflows, improving payment success rates, and delivering user-centric enhancements across e-commerce and POS payment surfaces.
- Designed and shipped an **installment-based Payment Plans system** supporting recurring, flexible patient payment schedules, with Celery Beat powering reliable asynchronous collection workflows — improving patient retention.
- Redesigned payment creation to be decoupled from webhook ordering, making the pipeline **idempotent and order-independent**, eliminating a class of race-condition and duplicate-payment failures.
- Implemented **application-level database sharding** to horizontally distribute data across multiple PostgreSQL instances, supporting sustained transaction growth without downtime.
- Introduced a **Redis caching layer** across 3+ microservices, reducing database load and improving API response latency for high-traffic billing and reporting endpoints, sustaining peak transaction surges without degradation.
- Provisioned and maintained AWS infrastructure (ECS, EC2, Lambda, SQS, RDS, API Gateway, Route 53, ECR, Load Balancers) across development and production, enabling **zero-downtime deployments**.
- Led major Python, SQLAlchemy, and pycogp2 version upgrades with zero regressions, reducing Docker image size by **25%**.
- Integrated third-party platforms (HubSpot, Vitally) to automate customer lifecycle tracking and surface usage analytics for customer success and business teams.
- Built **deposit reconciliation** workflows matching settled payments against bank deposits, cutting accounting errors.
- Built a **dispute-management module** with UI and email notifications, giving practice and admin users direct visibility into payment disputes.
- Hardened the public payment portal against an **active bot attack** by implementing Cloudflare Turnstile
- Implemented **AVS (Address Verification System)** checks and enforced a highly customisable maximum payment limit across all payment flows, strengthening platform-wide fraud controls.
- Mentored engineers across backend and QA teams on distributed system design, code review standards, and AWS best practices, consistently raising delivery velocity and engineering quality.
- Received the **Inspirational Mentor Award** and **Team Excellence Award** for cross-team mentorship and for leading the e-commerce/POS payment systems launch.

Software Engineer

08/2022 – 01/2023

Skit.ai — Voice AI SaaS for automated enterprise customer interactions

Bangalore

- Engineered performance and reliability improvements on a high-throughput Django backend serving **2M+ daily requests** between voice AI services and enterprise clients.
- Developed and validated proprietary ML-based Spoken Language Understanding (SLU) models using **100K+ labeled voice datasets**, contributing directly to model accuracy improvements.
- Collaborated cross-functionally to stabilize production services and optimize backend integrations for voice automation.

Software Engineer

09/2020 – 08/2022

Orange Business Services — Global IT, Cloud, IoT & Network Solutions Integrator

Gurugram

- Built a Django-based enterprise application for spend and subscription management, integrated with Microsoft SQL Server for multi-tenant data operations.
- Developed Python-based Azure Function microservices for an IoT sensor middleware platform, managing real-time and historical data streams, sensor configurations, and alarm systems for **critical water dam infrastructure**.
- Built real-time notification and alerting systems using Django ASGI, Channels, and WebSockets, delivering live updates via event-driven architecture across distributed clients.
- Built a Python CLI for automated log analysis, eliminating **400+ hours** of monthly manual investigation for L2 support.
- Developed text recognition and language conversion tools using Tesseract OCR and Google Translation APIs.
- **Above & Beyond Award** — recognized for exceptional contribution to the software incubation team.

Technical Skills

Languages	Python, SQL, Bash, Java, JavaScript
Backend	Django, Django REST Framework, Celery, Celery Beat, REST APIs, WebSockets, Spring Boot
Distributed Systems	Microservices, event-driven architecture, Kafka, RabbitMQ, sharding, caching, idempotency, webhooks
Data	PostgreSQL, Redis, Microsoft SQL Server, Redshift
Cloud & DevOps	AWS (EC2, ECS, Lambda, SQS, RDS, S3, API Gateway, Route 53, ELB), Azure Functions, Docker, CircleCI, GitLab CI/CD
AI Dev Tools	Claude Code, Cursor, Codex
AI / ML	OpenAI API, scikit-learn, H2O.ai

Education

M.Tech — Malaviya National Institute of Technology, Jaipur

07/2018 – 06/2020

B.Tech — University College of Engineering and Technology, Hazaribagh

07/2013 – 06/2017